

## Introduction

### Challenge:

Time series forecasting (TSF) is crucial for various application, such as weather forecasting, power load forecasting, and financial analysis. MLP-mixer-based models are the promising alternative to transformer-based models for TSF. However, the performance of the mixer-based models are still suboptimal.

### Solution:

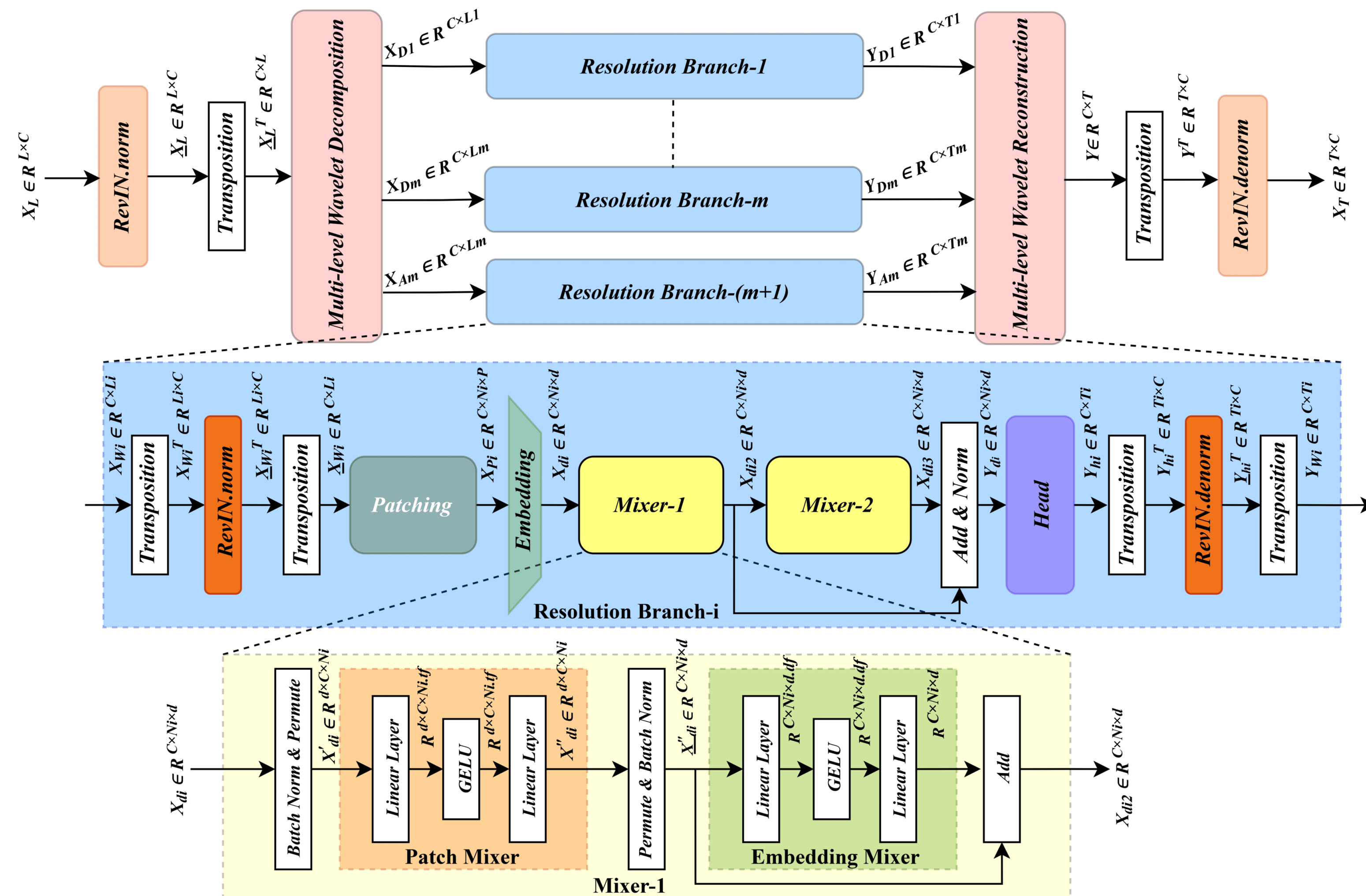
We propose **Wavelet Patch Mixer (WPMixer)**, a novel MLP-mixer-based model for long-term TSF. Three key components are,

1. **Multi-resolution wavelet decomposition** for extracting information in both frequency and time domains.
2. **Patching and embedding** to capture extended historical data and local information.
3. **MLP mixing** to incorporate global context.

### Performance:

WPMixer achieves reduction in **MSE by 2.2% to 7.8%** and in **MAE by 0.5% to 6.4%** compared to the-state-of-the-art model across ETT datasets. It is also **ten times** more computationally efficient than the state-of-the-art model.

## Model Architecture

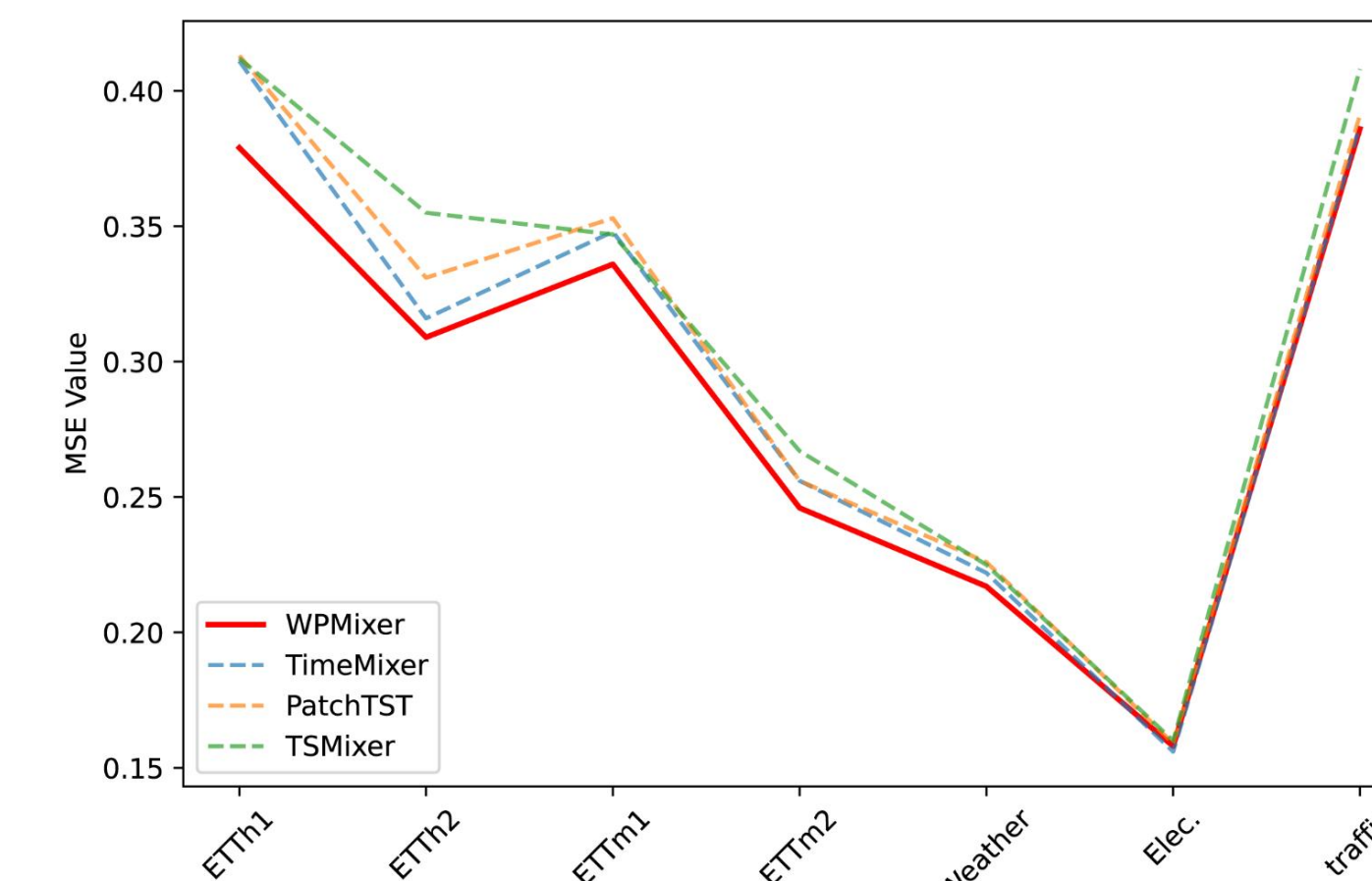


**Figure 1: WPMixer** with  $m$  levels of wavelet decomposition.  $X_{Ai}$  and  $X_{Di}$  are the approximation and detail coefficient series corresponding to the input time series  $X_L$ .  $Y_{Ai}$  and  $Y_{Di}$  are the predicted approximation and detail coefficient series corresponding to the predicted time series  $X_T$ .

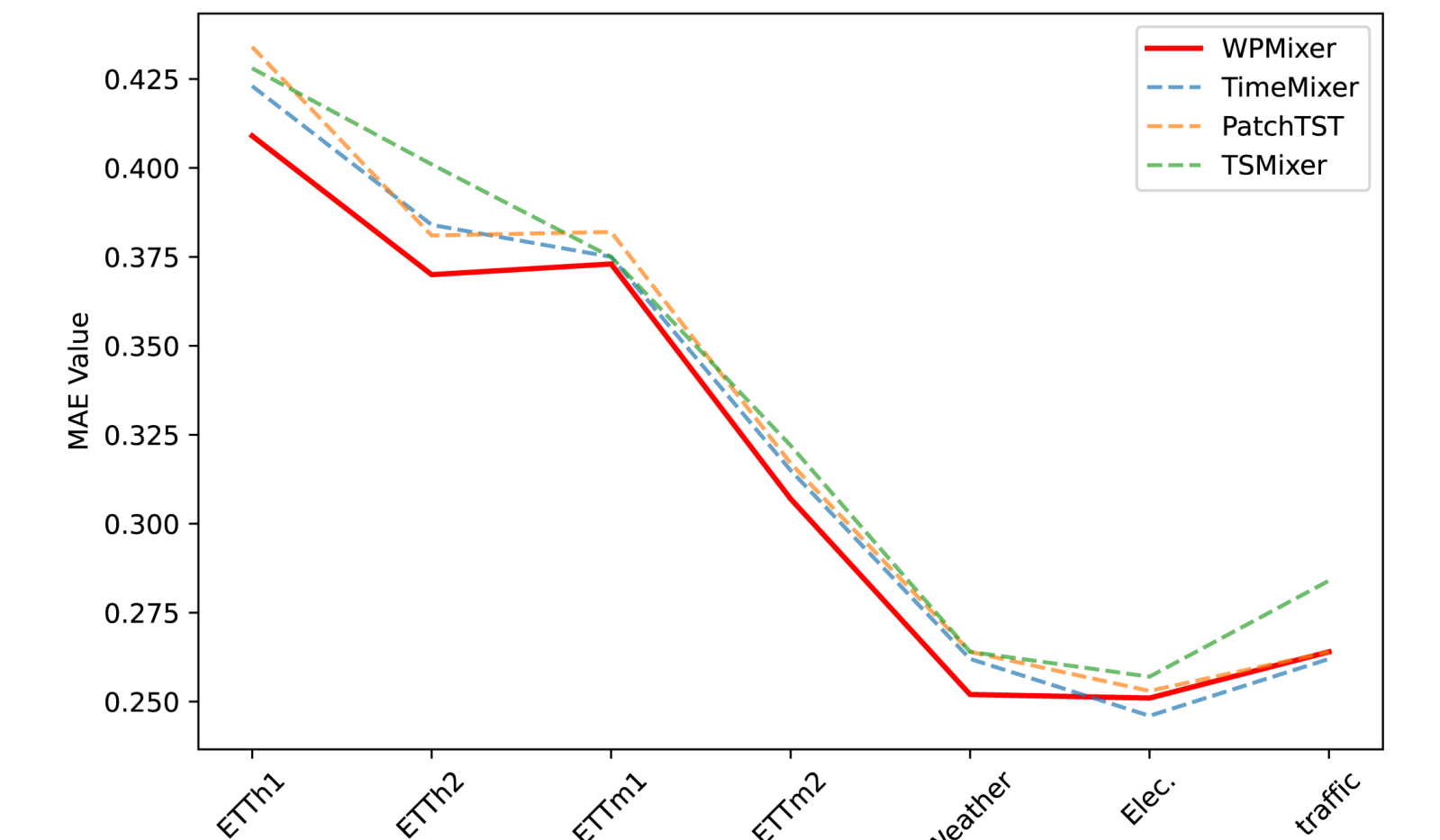
## Methodology

- Input time series window is decomposed into approximation and detail coefficient series using wavelet-based decomposition.
- Each coefficient series is analyzed individually using the resolution branch, which consists of patching, embedding and mixing across multi dimensions. Analyzing individually employing separate branch avoids information loss.
- Each branch predicts coefficient series. Final prediction is obtained by reconstructing the predicted coefficient series using the inverse wavelet transform.

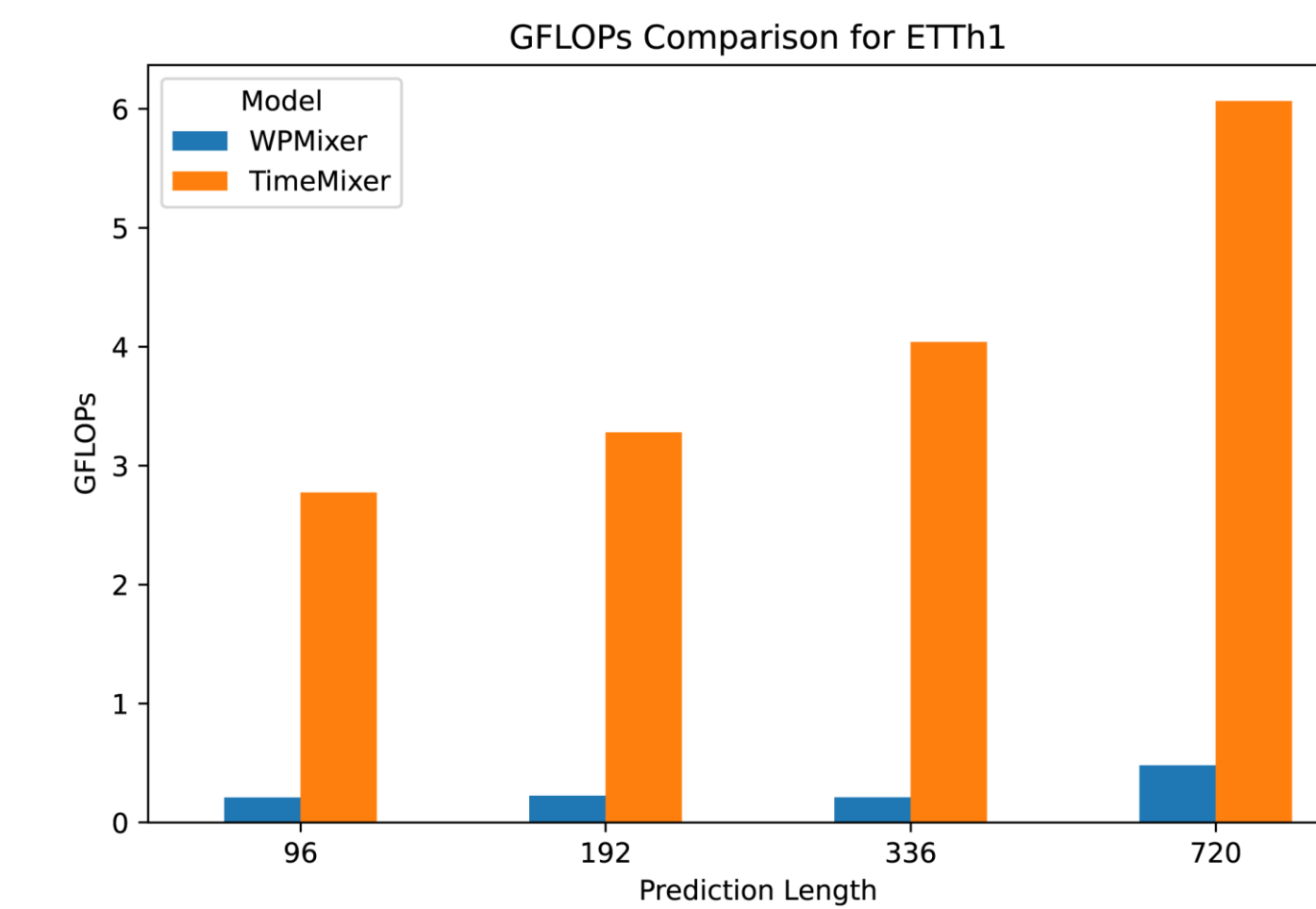
## Results



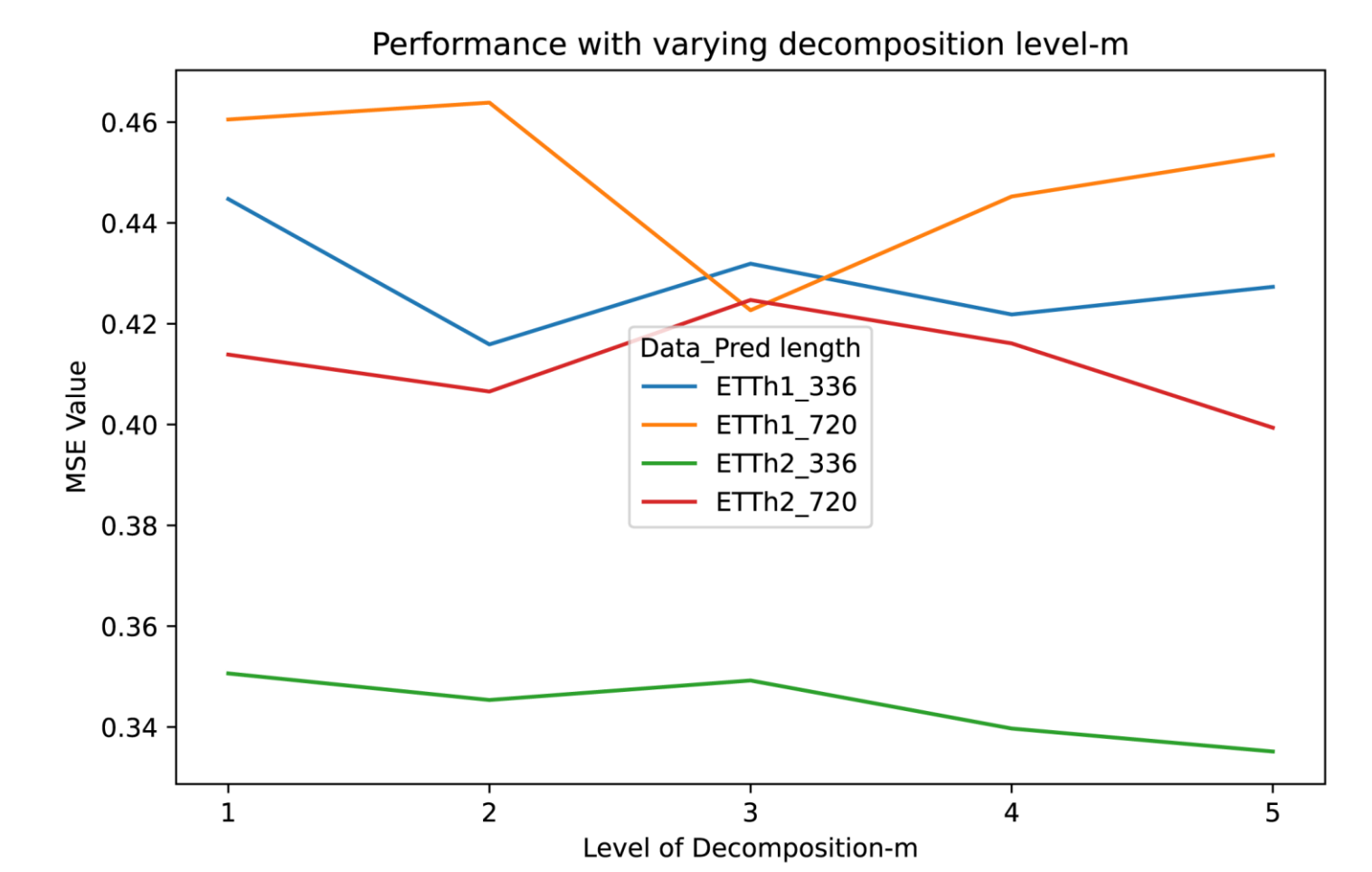
**Figure 2: Performance comparison using MSE.**



**Figure 3: Performance comparison using MAE.**



**Figure 4: Computational efficiency comparison.**



**Figure 5: Performance of WPMixer with varying  $m$ .**

## Conclusion

1. We evaluate the performance across two settings: fixed look-back window length and optimized look-back window length.
2. Our model achieves state-of-the-art performance across almost all datasets with optimal computational efficiency.
3. Performance varies with the level of decomposition. We searched for the optimum level of decomposition and the optimum wavelet type for each dataset.
4. The model's learning capability improves with increasing look-back window length. However, performance does not improve much after a certain length.